

Intra-die Spatial Correlation Extraction with Maximum Likelihood Estimation Method

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Abstract

In this paper, a novel intra-die spatial correlation extraction method is proposed. The proposed method is based on maximum likelihood estimation of multivariate normal distribution for multiple samples. The obtained likelihood function represents the underlying statistical relationships of all sample chips. The proposed method can account for the nugget effect due to white noise, and deals with deviations among samples caused by inter-die variation. Experimental results have shown that the proposed method is efficient and practical.

Key words: intra-die variations, spatial correlation, maximum likelihood estimation.

1 Introduction

As the minimum feature size of semiconductor device continues scaling down, integrated circuits suffer from increasing variations in the manufacturing process. These process variations lead to the geometric variations in the devices and interconnects, and greatly affect their electrical parameters. As a result, the performances of the fabricated circuits are degraded from the design specifications, and the manufacturing yield is lost. Therefore, it is more desirable to develop statistical analysis and design methodologies to tackle variation problems in the design stages [12].

Process variations can be classified into two categories according to the spatial scales. *Inter-die variations* affect device parameters with the same value on a die but with different values beyond the die scope, while *intra-die variations* cause device parameter values to vary across different locations within a single die. As technology generation marches, intra-die variations exceed inter-die variations and become predominant in total process variations. Intra-die variations often show spatial correlated patterns, which means devices that are closely placed tend to possess similar characteristics than those further apart.

Spatial correlation is defined to describe the degree to which the values of device characteristics are related to each other as a function of spatial distances of the devices [7]. Assumed to be a given function or matrix form, spatial correlation has been widely used in variation aware circuit analysis and design techniques, such as statistical timing analysis [3, 16], power/leakage minimization [2, 10]. Recently how to model spatial correlation from silicon measurement data has also attracted a lot of attention. The task of spatial correlation modeling [6–9, 11, 15] aims to extract the characteristic parameters of spatial correlation function provided with a large amount of silicon measurement data. It consists of two essential issues. Firstly, an appropriate kind of function form should be chosen to represent spatial correlation. The *positive semidefiniteness* property possessed by a valid spatial correlation function should be satisfied, which means that any correlation matrix generated from the correlation function is positive semidefinite. Secondly, the unknown parameters in the function should be extracted (or estimated) efficiently from measurement data, which contain unpredictable measurement error. In [11], the exponential function was used as spatial correlation function, and the parameter was extracted using the curve-fitting procedure. However, how to deal with measurement error in the extraction was not mentioned in [11]. In [15], the Matérn function was chosen to represent spatial correlation, and a least squares estimation (LSE) based method was presented to extract parameters of the correlation function. Moreover, measurement error was modeled as white noise. The method is robust enough to recover the correlation function even if the data are distorted by significant white noise. In [8], the unknown parameters of Matérn function were estimated by fitting the spectral domain counterpart of Matérn correlation function to the measurement data in spectral domain. The method can not only handle white noise measurement error, but also get rid of the influences of measurement error occurring in high frequency range. In [9], both exponential and Matérn functions were chosen as spatial correlation models. The maximum likelihood estimation (MLE) technique for a single test chip was proposed to find the required parameters to fit spatial correlation function to the measurement data of one chip. Therefore, a set of test chips will result in a set of correlation functions with different parameter values [9], which are hard to use in statistical analysis and design since only one correlation function is required. In addition, the method did not consider the purely random component and measurement error in measurement data, and the extraction results would be highly distorted.

In this paper, MLE-M (MLE for multiple test chips) method is proposed to obtain one spatial correlation function with a unique group of parameter values from a set of test chip measurement data. The main idea is to derive a single likelihood function for multiple chips by multiplying the set of likelihood functions for all test chips. The unknown parameters of the spatial correlation function are estimated from the single likelihood function for multiple chips. Moreover, in order to deal with the purely random component and measurement error in measurement data which are modeled as white noise, the spatial correlation function combined with the correlation of white noise is used to compute the

likelihood function for multiple chips. By this way, the spatial correlation function is recovered with high accuracy. Compared with the LSE based algorithm [15], the proposed MLE-M method has higher accuracy and efficiency as demonstrated by the experimental results.

The rest of this paper is organized as follows. In Section 2, we introduce the background about spatial correlation modeling based on random field theory. In Section 3, the MLE-M method for intra-die spatial correlation extraction is proposed. In Section 4, experimental results are presented. The paper is concluded in Section 5.

2 Background Material

According to [13], the intra-die variation Z can be further decomposed into three components:

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- a *deterministic* component Z_{det} , which is determined by layout context and can be modeled by deliberately exploring layout patterns;
- a *correlated random* (or *spatially correlated*) component Z_{cor} , which is random but shows correlated patterns due to proximity effects;
- a *purely random* component Z_{rnd} , which is spatially uncorrelated and can be treated as statistically random.

In this paper, for the sake of simplicity, the deterministic component is assumed to be well modeled and taken away from the whole process variation. We concentrate on the spatially correlated component, together with the purely random component and measurement error as [8, 15]. The spatially correlated component is generally modeled as random field in the literature. In this section, fundamental concepts and theories of random field are reviewed, and the Matérn correlation function is given. The purely random component and measurement error cause a discontinuity at the origin of the correlation function, which is called nugget effect. We will describe this phenomenon and give the modification form of the Matérn correlation function in this section.

2.1 Random Field [14]

Random field, also known as *stochastic process*, can be regarded as an indexed family of random variables denoted as $\{Z(\mathbf{s}) : \mathbf{s} \in D\}$, where D is a subset of d -dimensional Euclidean space $\mathbb{R}^d$. To specify a stochastic process, the joint probability distribution function of any finite subset $(Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_n))$ must be given in a consistent way, which is called *distribution* of the process. For ease of analysis, a random field is often assumed to be with *Gaussian* distribution, and is called Gaussian random field.

A random field has several key properties that are useful in practical problems. The field is *stationary* under translations, or *homogeneous*, if the distribution is unchanged when the point set is translated. The field is *isotropic* if the distribution is invariant under any rotation of the whole points in the parameter space. We study homogeneous isotropic field in this paper.

The *covariance* C and *correlation* R of a stochastic process are defined by

$$C(\mathbf{s}_i, \mathbf{s}_j) = \text{cov}(Z(\mathbf{s}_i), Z(\mathbf{s}_j)) = \text{E}[(Z(\mathbf{s}_i) - \text{E}[Z(\mathbf{s}_i)])(Z(\mathbf{s}_j) - \text{E}[Z(\mathbf{s}_j)])]$$

and

$$R(\mathbf{s}_i, \mathbf{s}_j) = C(\mathbf{s}_i, \mathbf{s}_j) / \sqrt{C(\mathbf{s}_i, \mathbf{s}_i)C(\mathbf{s}_j, \mathbf{s}_j)}$$

respectively for all $\mathbf{s}_i, \mathbf{s}_j \in D$, where $\text{E}[Z(\mathbf{s})]$ denotes the expectation of $Z(\mathbf{s})$. Thus a process is homogeneous if C and R depend only on the separation vector $\mathbf{h} = \mathbf{s}_i - \mathbf{s}_j$. Furthermore, it is isotropic if C and R depend upon $\mathbf{h}$ only through its length h , i.e.,

$$C(\mathbf{s}_i, \mathbf{s}_j) = C(\mathbf{h}) = C(h),$$

$$R(\mathbf{s}_i, \mathbf{s}_j) = R(\mathbf{h}) = R(h) = C(h)/C(0). \quad (1)$$

If we denote $C(0)$, the variance of $Z(\mathbf{s})$, as σ^2 , then the relationship between covariance and correlation is $C(h) = \sigma^2 R(h)$.

2.2 Correlation Function for The Spatially Correlated Component

The spatially correlated component is modeled as random field with variance σ^2 and correlation function $R(h)$ to be extracted. Positive semidefiniteness is the necessary condition for a parametric family of functions to define a legitimate class of correlation functions. However, this is not an easy condition to check directly. In practice, this is usually ensured by working within one of several standard classes of parametric model for $R(h)$. Although no certain criteria exist for spatial correlation representation, we prefer to select the Matérn function as it is more flexible than other correlation functions. However, the proposed extraction methods are also adaptive to other correlation functions. The Matérn function for a homogeneous isotropic field is in the form as

$$R(h) = \frac{1}{2^{\nu-1}\Gamma(\nu)}(\alpha h)^\nu K_\nu(\alpha h)$$

where K_ν is the second kind modified Bessel function of order ν , and Γ is the gamma function.

In Matérn function, α is the range parameter which measures how quickly correlation decays with distance, and ν is the shape parameter which controls

Figure 1: Matérn functions with different parameter groups.

Figure 2: Correlation function and spectral density of white noise.

smoothness of the process. When ν is small, it implies the field is rough. When ν is large, it means the field is smooth. Matérn function is a generalization of several other widely used correlation functions. For $\nu = 1/2$, Matérn function reduces to exponential function; when ν tends to infinity, it degenerates as Gaussian function. Fig. 1 shows Matérn functions with different parameter groups.

2.3 Correlation Function Considering Nugget Effect

In the measurement data, apart from the spatially correlated component, the purely random component and the unavoidable measurement error also exist. Typically, the two components are modeled as random variables with independent identical Gaussian distribution, or in short, Gaussian white noise. The correlation function of white noise and the corresponding spectral representation (named as “spectral density” [8]) are shown in Fig. 2.

When the two components are considered, the measurement data can still be regarded as a Gaussian random field, but the correlation function will have a discontinuity at the origin. This phenomenon is called “nugget effect” [5]. Denoting the sum variance of the above two components as τ^2 , the original Matérn correlation function $R(h)$ should be modified as

$$\tilde{R}(h) = \begin{cases} 1 & \text{if } h = 0 \\ \frac{\sigma^2}{\sigma^2 + \tau^2} \cdot R(h) = \frac{\sigma^2}{\sigma^2 + \tau^2} \cdot \frac{1}{2^{\nu-1}\Gamma(\nu)} (\alpha h)^\nu K_\nu(\alpha h) & \text{if } h > 0 \end{cases}$$

Fig. 3 shows the modified Matérn correlation functions considering the nugget effect for the original Matérn correlation functions in Fig. 1.

In [9], $R(h)$ was simply used in the extraction. In this paper, the proposed method uses the modified form $\tilde{R}(h)$ to account for the nugget effect and the extraction results are more accurate.

Figure 3: Modified Matérn functions with different parameter groups.

3 Intra-die Spatial Correlation Extraction Based on MLE Method

In this section, we will first formulate the problem of spatial correlation function extraction. We further deduce the likelihood function for multiple samples, and present the extraction method in details.

3.1 Problem Formulation

In the measurement process, we sample to gather measurement data over a batch of M chips, with each chip comprising N measurement sites. The spatial locations of the measurement sites on each chip should be identical.

Assume that the spatially correlated component of intra-die variation is modeled as a Gaussian random field with Matérn correlation function $R(\vec{\psi}; h)$ (denoted as $R(\vec{\psi})$ for short). The problem of spatial correlation function extraction is defined to be: given measurement data representing process variations, extract the unknown parameter vector $\vec{\psi} = (\alpha, \nu)$ that completely specifies the correlation function, and the variance of the spatially correlated component σ^2 , while the interference of nugget effect and inter-die variation should be considered. The objective is to recover the spatial correlation function $R(\vec{\psi})$ as accurately as possible after plugging the estimated $\vec{\psi}$ into Matérn function.

3.2 Spatial Correlation Extraction Based on MLE for Multiple Samples

For a sample $\vec{z} = (z(\mathbf{s}_1), \dots, z(\mathbf{s}_N))^T$ from an N -variate Gaussian random field with zero mean vector and $N \times N$ covariance matrix $C = \sigma^2 R(\vec{\psi})$, the distribution is

$$p(\vec{z} | \sigma^2, \vec{\psi}) = \frac{1}{(2\pi)^{N/2} (\det(\sigma^2 R))^{1/2}} \cdot \exp \left(-\frac{1}{2\sigma^2} \vec{z}^T R^{-1} \vec{z} \right)$$

Likelihood function is defined by reversing the roles of the data vector and the parameter vector in the distribution, as $L(\sigma^2, \vec{\psi} | \vec{z})$, to represent the likelihood of the parameters given the observed data, and $L(\sigma^2, \vec{\psi})$ is often used with the understanding that the observed data $\vec{z}$ is implicit. In practice, for computational convenience the logarithm of the likelihood is taken and estimates are obtained by maximizing the log-likelihood function. The log-likelihood function of a sample is

$$\log L(\sigma^2, \vec{\psi}) = -\frac{N}{2} \log 2\pi - \frac{N}{2} \log \sigma^2 - \frac{1}{2} \log \det R - \frac{1}{2\sigma^2} \vec{z}^T R^{-1} \vec{z} \quad (2)$$

As there are M samples obtained in the measurement process, in [9], Eq. 2 is used M times to estimate the unknown parameters of each sample separately, resulting in M groups of different parameter values unrelated to each other.

However, designers do not simply care for the parameters of each individual chip. Except for the particular characteristic of each specific chip in these samples, there is no confidence for which group of parameter values to choose for inference. The method in [9] provides no guidance for robust circuit design, where we actually need the unique group of parameters that capture the characteristics buried in all samples.

In fact since we have M samples of observations $\vec{z}_1, \dots, \vec{z}_M$, with the m -th sample containing N variates as $\vec{z}_m = (z_m(\mathbf{s}_1), \dots, z_m(\mathbf{s}_N))^T$, it is more natural to calculate the likelihood based upon all samples, i.e., we want to search for the parameters that maximize the likelihood of all the samples. Before we do that, two processing steps should be performed to coincide with the actual extraction process. Firstly, as the M samples of observations are from M different chips, they are inevitably affected by inter-die variations. Although the percentage of inter-die variations in total process variations becomes comparatively smaller in the sub-micron regime, the extraction results will be distorted if the influence is neglected. To alleviate the sample deviations caused by inter-die variations, we remove the mean effect through a simple averaging process as

$$z_m^*(\mathbf{s}_i) = z_m(\mathbf{s}_i) - \frac{1}{M} \sum_{k=1}^M z_k(\mathbf{s}_i)$$

and the m -th sample of data vector becomes $\vec{z}_m^* = (z_m^*(\mathbf{s}_1), \dots, z_m^*(\mathbf{s}_N))^T$. Secondly, as mentioned in Sect. 2.3, the purely random component and measurement error result in the nugget effect, which should be taken into consideration in the extraction. Replacing $R(h)$ with $\tilde{R}(h)$, the modified covariance matrix of measurement data becomes

$$\tilde{C} = \sigma^2 R(\vec{\psi}) + \tau^2 I$$

where $\sigma^2 R(\vec{\psi})$ is the covariance of the spatially correlated component, while $\tau^2 I$ is the covariance caused by nugget effect. Parameterizing

$$\kappa = \tau^2 / \sigma^2,$$

the modified correlation matrix of measurement data is

$$\tilde{R}(\kappa, \vec{\psi}) = R(\vec{\psi}) + \kappa I.$$

The N -variate Gaussian measurement process can be denoted as $Z \sim N(\mathbf{0}, \sigma^2 \tilde{R}(\kappa, \vec{\psi}))$.

After the processings, the likelihood function for all the M samples is [1]

$$L(\sigma^2, \kappa, \vec{\psi}) = \prod_{m=1}^M L_m(\sigma^2, \kappa, \vec{\psi}) = \frac{1}{(2\pi)^{MN/2} (\det(\sigma^2 \tilde{R}))^{M/2}} \cdot \exp \left(-\frac{1}{2\sigma^2} \sum_{m=1}^M \vec{z}_m^{*T} \tilde{R}^{-1} \vec{z}_m^* \right)$$

and the log-likelihood function is

$$\log L(\sigma^2, \kappa, \vec{\psi}) = \log \left(\prod_{m=1}^M L_m(\sigma^2, \kappa, \vec{\psi}) \right) = -\frac{MN}{2} \log 2\pi - \frac{MN}{2} \log \sigma^2 - \frac{M}{2} \log \det \tilde{R} - \frac{1}{2\sigma^2} \sum_{m=1}^M \vec{z}_m^{*T} \tilde{R}^{-1} \vec{z}_m^* \quad (3)$$

By setting $\frac{\partial \log L}{\partial \sigma^2} = 0$, we can get the estimation of σ^2 as

$$\hat{\sigma}^2(\tilde{R}) = \frac{1}{MN} \sum_{m=1}^M \vec{z}_m^{*T} \tilde{R}^{-1} \vec{z}_m^* \quad (4)$$

Substituting Eq. 4 back into Eq. 3 and ignoring the constants, we can obtain the concentrated log-likelihood function

$$\log L_0(\kappa, \vec{\psi}) = -\log \det \tilde{R} - N \log \hat{\sigma}^2(\tilde{R}) \quad (5)$$

Regard the scalar $\vec{z}_m^{*T} \tilde{R}^{-1} \vec{z}_m^*$ as the trace of a 1×1 matrix. Using the property of the trace of a matrix that $\text{tr}(AB) = \text{tr}(BA)$ whenever A and B are matrices so shaped that both products exist, we can reformulate $\hat{\sigma}^2$ as

$$\hat{\sigma}^2 = \frac{1}{MN} \sum_{m=1}^M \text{tr}(\vec{z}_m^{*T} \tilde{R}^{-1} \vec{z}_m^*) = \frac{1}{MN} \sum_{m=1}^M \text{tr}(\vec{z}_m^* \vec{z}_m^{*T} \tilde{R}^{-1}) = \frac{1}{N} \text{tr} \left(\frac{1}{M} \sum_{m=1}^M \vec{z}_m^* \vec{z}_m^{*T} \tilde{R}^{-1} \right) = \frac{1}{N} \text{tr}(Y \tilde{R}^{-1}) \quad (6)$$

where

$$Y = \frac{1}{M} \sum_{m=1}^M \vec{z}_m^* \vec{z}_m^{*T}$$

Substituting Eq. 6 back into Eq. 5 and ignoring the constant, we obtain the log-likelihood function of MLE for multiple samples as

$$\log L_0(\kappa, \vec{\psi}) = -\log \det \tilde{R} - N \log(\text{tr}(Y \tilde{R}^{-1}))$$

This must be optimized numerically with respect to κ and $\vec{\psi}$, followed by back substitution to obtain $\hat{\sigma}^2$.

In the numerical evaluation steps of log-likelihood function, we find that direct computation of the log determinant of $\tilde{R}$ often incurs "log of zero" error, as the determinant of $\tilde{R}$ may approach to zero too closely. However, if there is no numerical errors, a positive value close to zero will result in a negative value in the acceptable range after taking the logarithm. To overcome this problem, we first decompose $\tilde{R}$ into two matrices L and U using LU factorization, where U is an upper triangular matrix, and L is the permutation of a lower triangular matrix whose diagonal entries are all ones. Since $\tilde{R}$ should be positive definite, $\det \tilde{R} > 0$. Using the property of the determinant that $\det \tilde{R} = |\det L| \cdot |\det U|$,

and $|\det L| = 1$, the log determinant of $\tilde{R}$ can be obtained by summing up the absolute values of the diagonal elements of the U matrix, i.e.,

$$\log \det \tilde{R} = \log |\det U| = \log \left| \prod_{j=1}^N u_{jj} \right| = \sum_{j=1}^N \log |u_{jj}|$$

where u_{mm} is the diagonal element of U and $|\bullet|$ means taking the absolute value. The finally obtained log-likelihood function is in the form as

$$\log L_0(\kappa, \vec{\psi}) = - \sum_{j=1}^N \log |u_{jj}| - N \log(\text{tr}(Y \tilde{R}^{-1}))$$

This can be solved by any standard nonlinear optimization technique. In our implementation, we use the *fmincon* function in MATLAB which is based on a sequential quadratic programming method.

4 Experimental Results

The proposed method was implemented in MATLAB on an Intel machine with 3.0 GHz XEON CPU. Without real silicon measurement data, we synthesized a pseudo measurement process and attained data from a batch of M chips with N sites on each chip. The area of the chip is set to be $10 \text{ mm} \times 10 \text{ mm}$. We conducted two groups of experiments according to different sampling schemes. In the first group, the N sites are evenly distributed on the chip in horizontal and vertical directions respectively, which is called “uniform gridding sampling scheme”. In the second group, the N sites reside on the two-dimensional plane in a completely random way, which is called “Monte Carlo sampling scheme”. The synthetic data consist of four components, i.e., the spatially correlated component of intra-die variation, the purely random component of intra-die variation, inter-die variation, and measurement error. The spatially correlated component is the most important part, which is generated as a homogeneous isotropic field with Gaussian distribution and Matérn correlation function. The “Cholesky decomposition” method described in [4] and used in [8, 9] was also utilized for the generation. Fig. 4 shows the surface plot of a generated random field. The purely random component and measurement error were added as Gaussian white noise, whose variances were chosen with different percentages (labeled as κ) w.r.t. the variance of the spatially correlated component. For inter-die variation, a global variation component was added to the characteristic values of all sites on a same chip. Different chips were added with different global variation values from a Gaussian distribution. The above three variation components were generated with a set of different variance values. Fig. 5 shows the surface plot of the finally generated measurement data from a sample chip.

We implemented the proposed method in two branches for comparison. One does not consider nugget effect and use the original $R(h)$ in likelihood function

Figure 4: Random field.

Figure 5: Measurement data.

for estimation which is called MLEsim, while the other accounts for nugget effect and use the modified form $\hat{R}(h)$ in likelihood function which is called MLEnug. We also implemented the LSE based extraction algorithm in [15] named as RESCF. We performed multi-runs (30 times) in each experiment and present the average results in Tables~1 and~2.

In the experiments, the relative error of the variance of the spatially correlated component was given by

$$\text{err}(\sigma^2) = \left| \frac{\hat{\sigma}^2 - \sigma^2}{\sigma^2} \right|,$$

and the relative error of the spatial correlation function was computed by

$$\text{err}(R(h)) = \frac{\|\hat{R}(h) - R(h)\|}{\|R(h)\|},$$

in which σ^2 and $R(h)$ denote the actual variance and spatial correlation function used to generate the random field, while $\hat{\sigma}^2$ and $\hat{R}(h)$ refer to the extracted variance and spatial correlation function after plugging in the estimated parameters.

From the two tables, we can see that when nugget effect is ignored in the extraction, MLEsim works badly. For most of the test cases, the relative errors of the extracted spatial correlation function are above 30%, or even higher to 70% which are totally unacceptable. When nugget effect is considered, MLEnug works well, and achieves better results than RESCF for all test cases with less runtime for both two sampling schemes. This shows the accuracy and efficiency of the proposed method. Fig. 6 illustrates the correlation functions extracted by the three methods compared with the actual correlation function in one experiment.

Table 1: Results of spatial correlation extraction with uniform gridding sampling scheme

M	N	κ	RESCF			MLEsim			MLEnug		
			$\text{err}(\sigma^2)$	$\text{err}(R(h))$	$t(\text{sec.})$	$\text{err}(\sigma^2)$	$\text{err}(R(h))$	$t(\text{sec.})$	$\text{err}(\sigma^2)$	$\text{err}(R(h))$	$t(\text{sec.})$
500	11×11	10%	2.14%	1.69%	10.43	10.98%	5.50%	2.73	0.95%	0.58%	7.63
		50%	6.05%	3.81%	12.33	50.21%	23.83%	2.04	1.95%	1.17%	5.48
		100%	7.31%	4.26%	10.31	98.76%	38.69%	2.80	4.12%	2.27%	5.89
1000	11×11	10%	2.45%	1.58%	9.62	11.66%	5.44%	2.60	0.84%	0.45%	7.85
		50%	4.94%	3.02%	15.23	50.64%	23.71%	2.14	2.17%	1.02%	5.91
		100%	6.58%	3.90%	12.24	99.18%	38.57%	2.78	3.18%	1.75%	5.97
500	21×21	10%	1.19%	1.09%	137.77	13.16%	13.90%	30.72	0.38%	0.30%	131.28
		50%	1.65%	1.66%	231.66	25.93%	42.41%	34.15	0.99%	0.63%	87.84
		100%	2.75%	1.98%	227.44	79.64%	55.72%	54.56	1.79%	0.77%	70.38
1000	21×21	10%	0.95%	0.75%	185.31	12.90%	13.88%	30.72	0.30%	0.27%	132.76
		50%	1.69%	1.32%	238.66	26.09%	42.42%	34.82	1.07%	0.48%	90.48
		100%	2.46%	1.60%	222.42	80.04%	55.66%	55.32	1.37%	0.68%	68.15

Table 2: Results of spatial correlation extraction with Monte Carlo sampling scheme

M	N	κ	RESCF			MLEsim			MLEnug		
			err(σ^2)	err($R(h)$)	$t(\text{sec.})$	err(σ^2)	err($R(h)$)	$t(\text{sec.})$	err(σ^2)	err($R(h)$)	$t(\text{sec.})$
500	11×11	10%	1.85%	2.20%	15.93	25.53%	45.34%	2.27	0.63%	0.70%	7.55
		50%	2.01%	2.33%	16.65	27.80%	63.16%	2.57	1.59%	1.01%	5.14
		100%	2.32%	2.18%	14.98	84.23%	71.75%	2.50	1.67%	1.39%	5.00
1000	11×11	10%	1.63%	1.65%	15.04	23.69%	42.48%	2.17	0.35%	0.45%	7.41
		50%	1.57%	1.30%	15.50	27.48%	63.49%	2.50	1.29%	0.91%	5.19
		100%	1.93%	1.52%	15.03	86.13%	72.93%	2.19	1.49%	1.15%	4.93
500	21×21	10%	1.83%	1.27%	200.43	47.15%	56.20%	32.88	0.39%	0.32%	109.78
		50%	1.71%	1.33%	217.19	7.58%	72.74%	32.81	1.34%	0.60%	84.32
		100%	1.92%	1.36%	177.16	70.16%	77.92%	25.16	1.32%	0.67%	79.22
1000	21×21	10%	1.33%	1.20%	209.48	46.04%	53.49%	34.27	0.32%	0.29%	107.84
		50%	1.12%	1.06%	223.88	8.70%	71.51%	33.83	0.80%	0.54%	84.05
		100%	0.91%	1.14%	204.29	69.44%	77.61%	27.86	0.81%	0.47%	79.96

Figure 6: Comparison of correlation function extraction.

5 Conclusion Remarks

Intra-die spatial correlation extraction has been a prevailing subject these years. In this paper, we propose a novel extraction method using MLE technique. The proposed method is based on MLE of multivariate normal distribution for multiple samples. After removing the sample deviations caused by inter-die variations, the proposed method takes all sample chips as a whole to formulate the likelihood function, in which nugget effect is integrated to account for the purely random component of intra-die variation with measurement error. Experimental results show that the proposed method outperforms the LSE based extraction algorithm [15] for all test cases with less runtime. The proposed method is more efficient and stable for spatial correlation extraction provided with real silicon data.

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